Reward Learning

Dylan Losey


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TURBO
MORE GAMES

This Lecture

- Learning and optimizing rewards in robotics
- What is inverse reinforcement learning?
- Theory of mind


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Multi-Armed Bandit

A standard **multi-armed bandit** is defined by:

- A set of K different arms we can pull (i.e., actions we can take)
- Unknown reward parameters $\theta = (\theta_1, \ldots \theta_K)$

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- Unknown reward parameters $\theta = (\theta_1, \dots \theta_K)$

At each interaction:

- We pull an arm $a \in (1, \dots, K)$ of our choice
- Get reward $R(a) = 1$ with probability θ_a otherwise get reward $R(a) = 0$
- $\mathrm{E}[R(a)] = 1 \cdot \theta_a + 0 \cdot (1 - \theta_a) = \theta_a$

Multi-Armed Bandit

The **regret** after T total interactions is:

$$\rho = T \max \theta - \sum_{t=1}^{T} r^t$$

Expected reward for
best arm every interaction

Sum of rewards we received

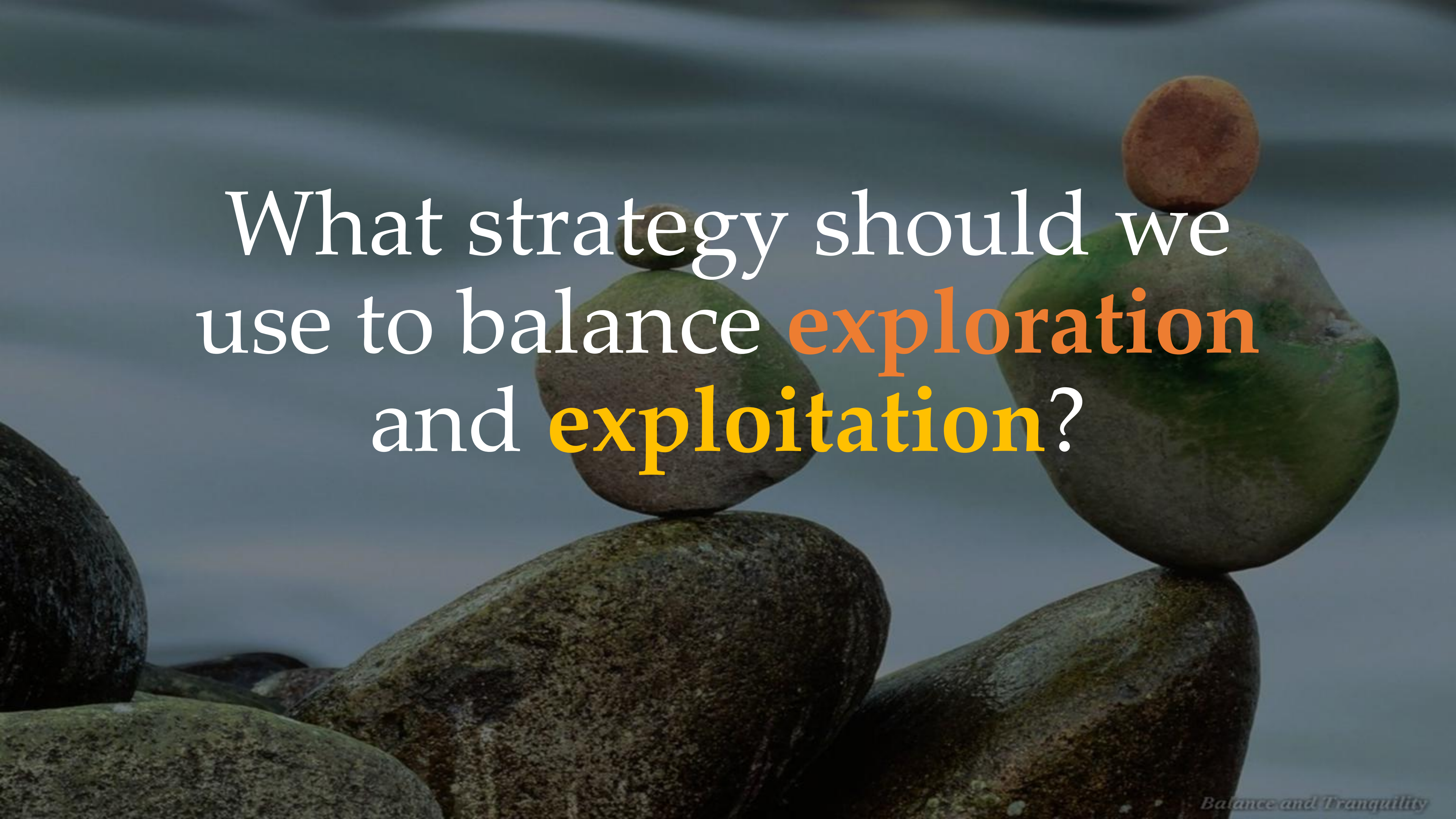
		1
0		
	1	
		0
	1	

Exploration vs. Exploitation

Exploration.
Pull each arm enough time to estimate θ

Exploitation.
Pull the best arm to minimize regret

Trade-off.
How do we balance between exploring and exploiting?



What strategy should we use to balance **exploration** and **exploitation**?

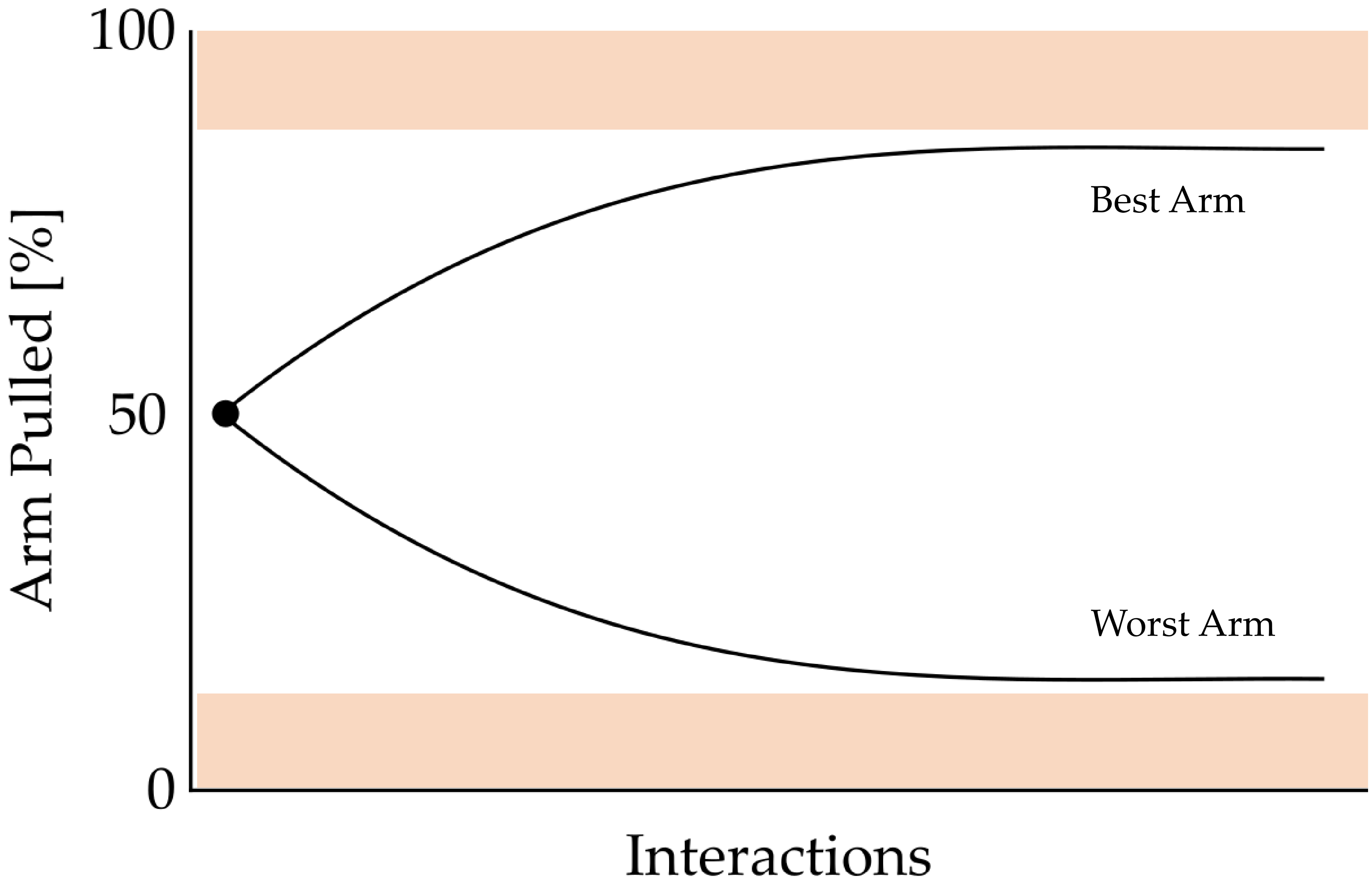
Epsilon-Greedy Strategy

At each interaction:

- $\mu \leftarrow (\mu_1, \ldots \mu_K)$ the average reward for each arm so far
- $p \leftarrow rand(0, 1)$ random number between 0 and 1
- If $p < \epsilon$:

  $a \leftarrow random\_arm$
  Explore. Pull arm uniformly at random
- Else:

  $a \leftarrow \text{argmax}\, \mu$
  Exploit. Pull arm with maximum average reward


100
50
0
Arm Pulled [%]
Best Arm
Worst Arm
Interactions

What is inverse reinforcement learning?

Guided Cost Learning:

Deep Inverse Optimal Control via Policy Optimization

Chelsea Finn, Sergey Levine, Pieter Abbeel
UC Berkeley

Inverse Reinforcement Learning

A **Markov decision process** (**MDP**) is a tuple:

$$\mathcal{M} = \langle S, A, f, r, \gamma \rangle$$

- $s \in S$ are states
- $a \in A$ are actions
- $f(\mathrm{s}, \mathrm{a})$ or $f(s^{t+1} \mid s^t, a^t)$ are dynamics
- $r(s)$ is the reward function
- γ is the discount factor

Inverse Reinforcement Learning

A **Markov decision process** (**MDP**) is a tuple:

$$\mathcal{M} = \langle S, A, f, r, \gamma \rangle$$

Given an MDP, we can find a robot policy that maximizes reward. We have covered (and will cover) methods to solve this:

- Value iteration
- Reinforcement learning

Inverse Reinforcement Learning

In **inverse reinforcement learning (IRL)** we have MDP *without rewards*:

$$\mathcal{M} = \langle S, A, f, \gamma \rangle$$

Inverse Reinforcement Learning

In **inverse reinforcement learning** **(IRL)** we have MDP *without rewards*:

$$\mathcal{M} = \langle S, A, f, \gamma \rangle$$

Instead of the reward function, we are given demonstrations:

$$D = \left((s^1, a^1), (s^2, a^2) \ldots (s^N, a^N) \right)$$

Inverse Reinforcement Learning

In **inverse reinforcement learning** **(IRL)** we have MDP *without rewards*:

$$\mathcal{M} = \langle S, A, f, \gamma \rangle$$

$$D = \left((s^1, a^1), (s^2, a^2) \ldots (s^N, a^N) \right)$$

Our first step is *recovering* the reward function from the demonstrations.

What is the **reward function?**

Rewards

Let the reward function have parameters θ.

The robot knows the rest of the reward, but does not know the right θ.

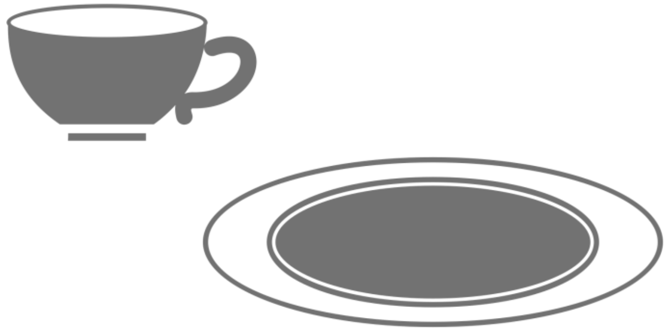
Rewards

Let the reward function have parameters θ.

The robot knows the rest of the reward, but does not know the right θ.

Example: Should we reach the cup or plate?

$$r(s, \boldsymbol{\theta}) = \boldsymbol{\theta} \cdot \begin{bmatrix} -\mathrm{dist}(s, \mathrm{cup}) \\ -\mathrm{dist}(s, \mathrm{plate}) \end{bmatrix}$$



Rewards

Let the reward function have parameters θ.

The robot knows the rest of the reward, but does not know the right θ.

Example: Should we reach the cup or plate?

$$r(s, \boldsymbol{\theta}) = \underline{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} \cdot \begin{bmatrix} -\mathrm{dist}(s, \mathrm{cup}) \\ -\mathrm{dist}(s, \mathrm{plate}) \end{bmatrix}$$

For this θ the robot should go directly to cup

Rewards

Let the reward function have parameters θ.

The robot knows the rest of the reward, but does not know the right θ.

In **inverse reinforcement learning** we write the reward function as:

$$r(s, \theta)$$

Reward function

Unknown parameters θ

State

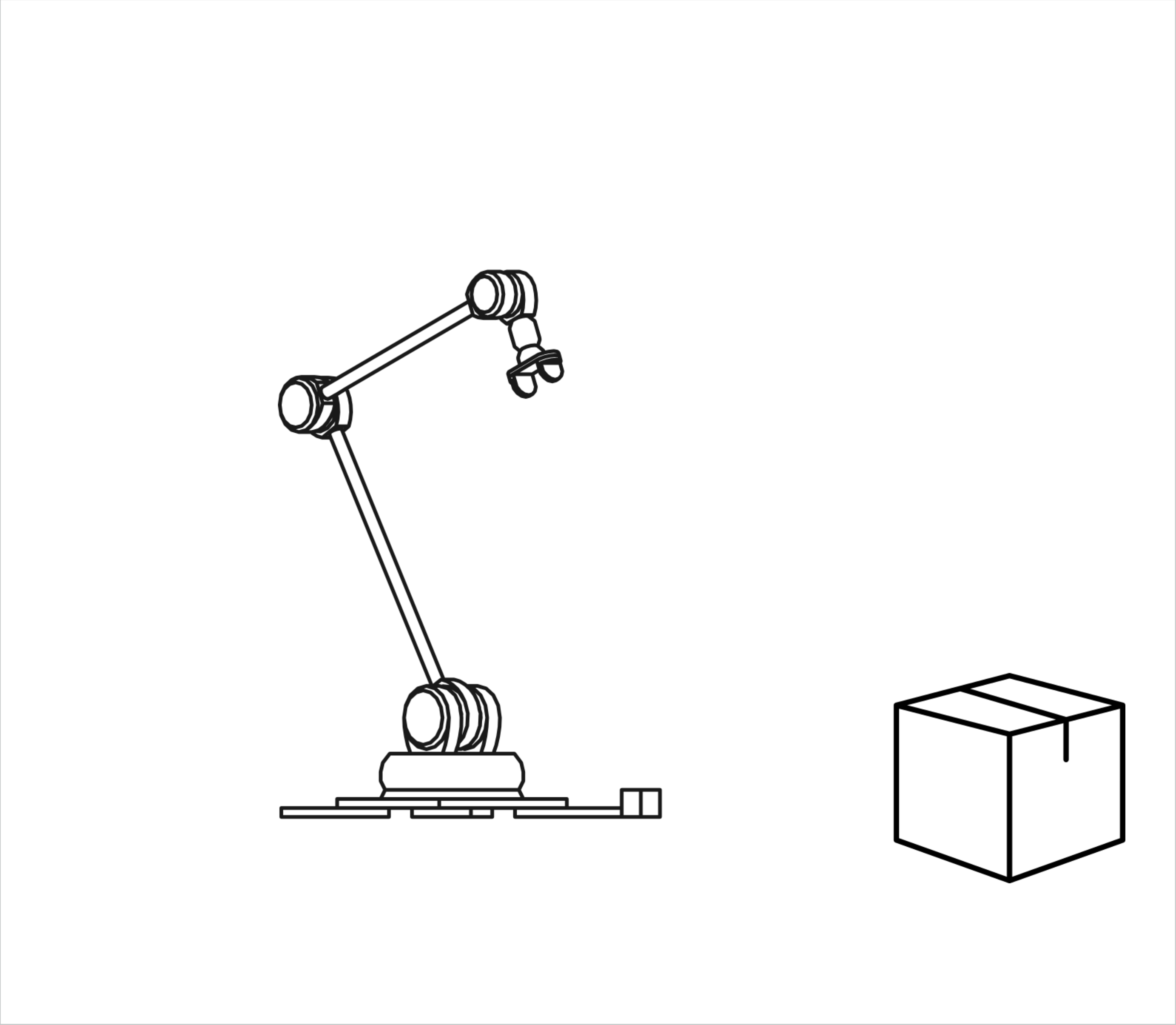
Inverse Reinforcement Learning

Robot is operating in **environment**:

$$\mathcal{M} = \langle S, A, f, \gamma \rangle$$

MDP without rewards

Robot does not know what it should be trying to do.

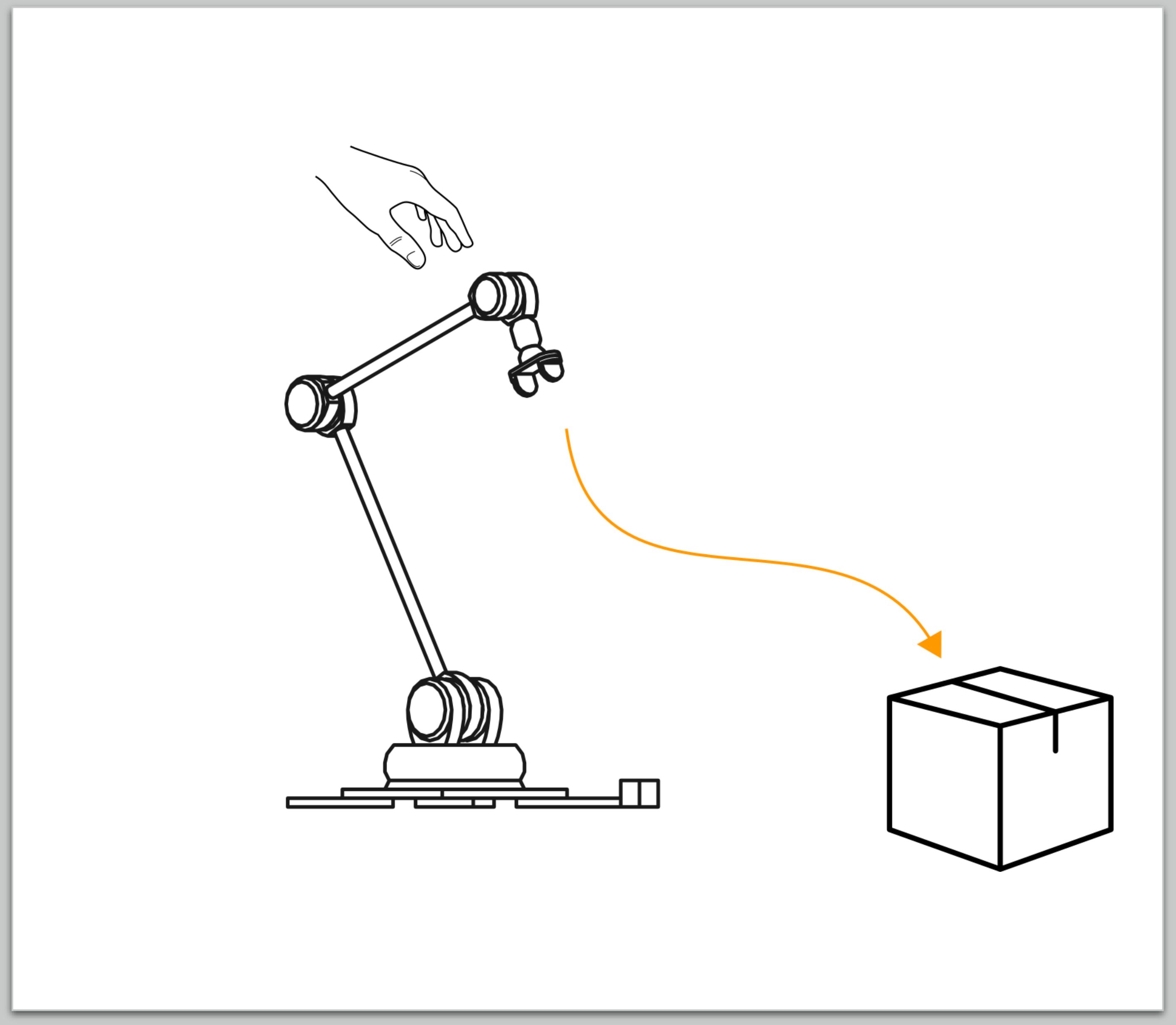


Inverse Reinforcement Learning

Human expert provides **demonstrations**.

$$D = \left((s^1, a^1), \dots (s^N, a^N) \right)$$

Expert state-action pairs



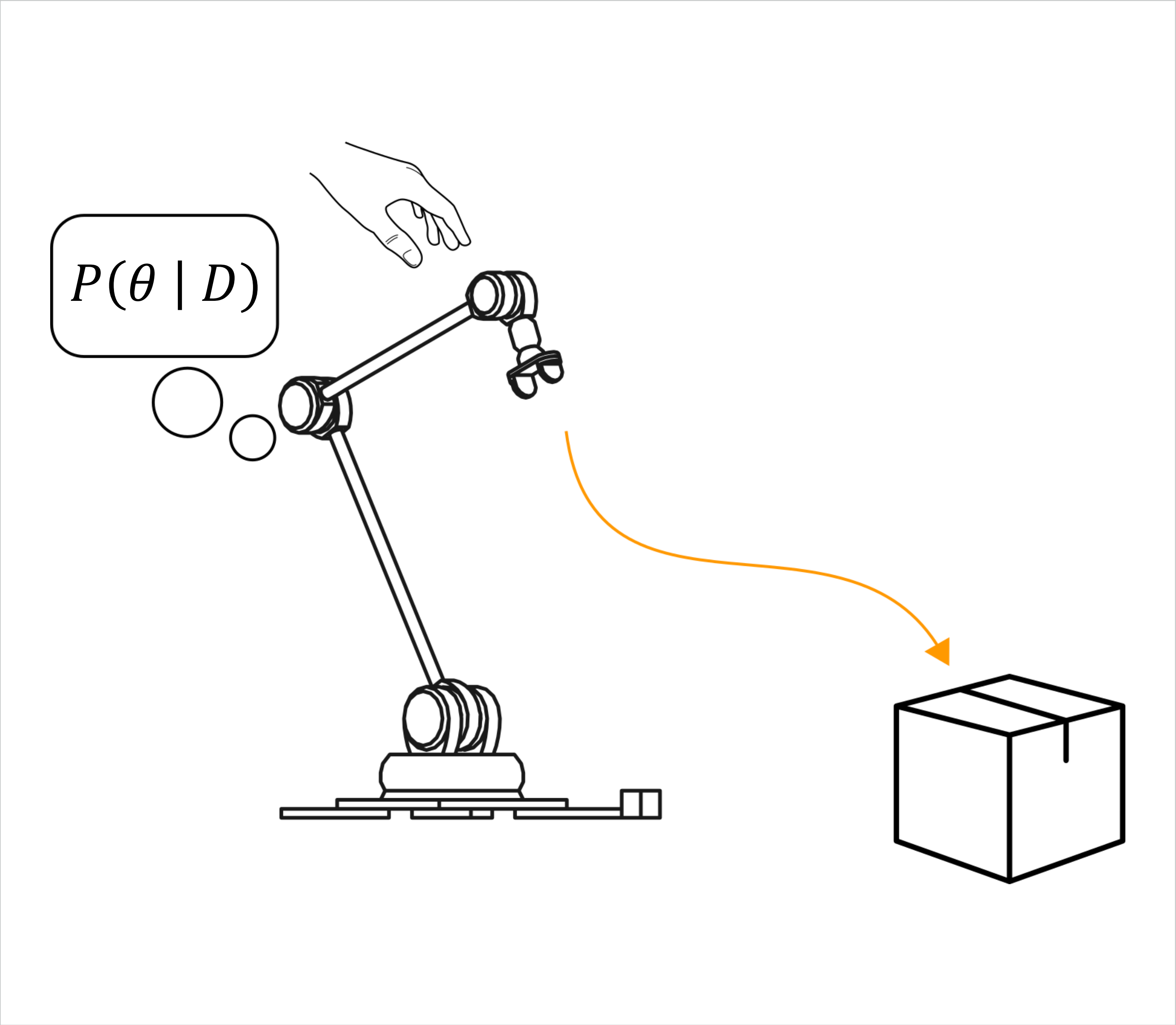
Inverse Reinforcement Learning

1. Robot **infers what reward** the human was optimizing

$$r(s, \theta)$$

Reward function

Unknown parameters θ

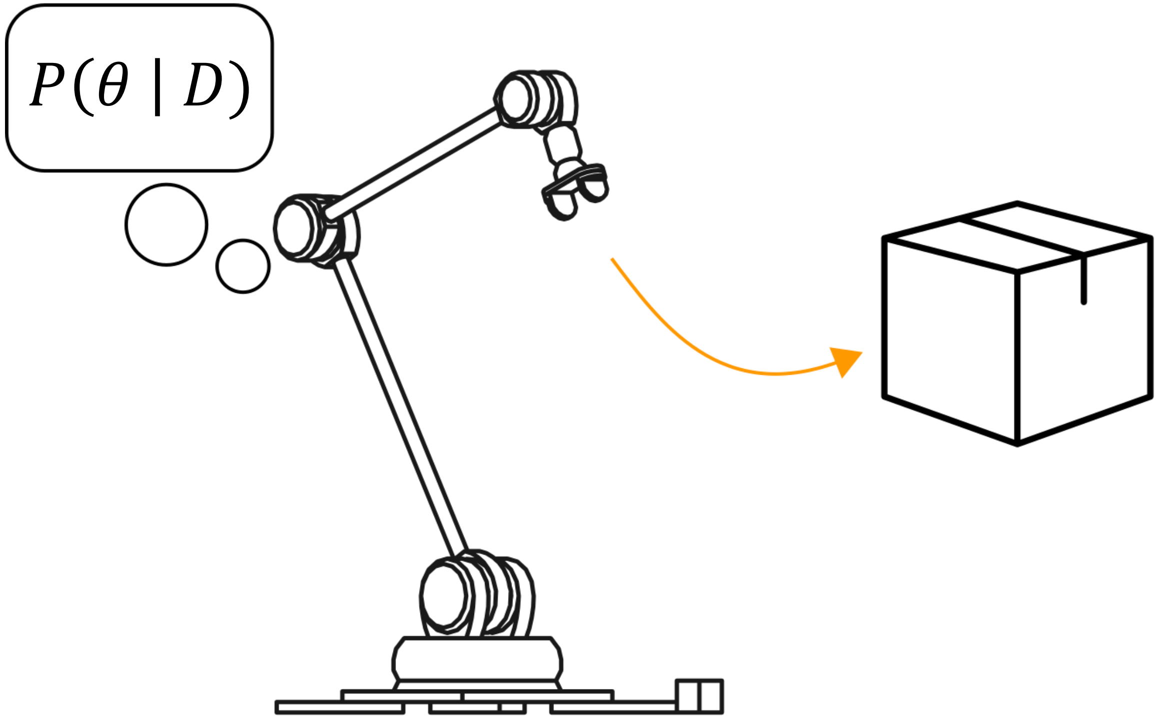


Inverse Reinforcement Learning

2. Robot **optimizes for** the learned reward

$$\mathcal{M} = \langle S, A, f, r, \gamma \rangle$$

MDP with learned reward



Challenges

1. How do we infer a human's reward from demonstrations?
   - Bayes theorem to find $P(\theta \mid D)$
2. How do we optimize over the learned reward?
   - Reinforcement learning

Sampling

$$P(\theta \mid D) \propto P(\theta) \prod_{(s,a) \in D} P(a|s,\theta)$$

For now lets assume we only have one state-action pair

$$P(\theta \mid D) \propto P(a|s,\theta) \cdot P(\theta)$$

Sampling

$$P(\theta \mid D) = \frac{P(a|s,\theta) \cdot P(\theta)}{P(a \mid s)}$$

Bayes theorem where we assume $P(\theta) = P(\theta \mid s)$

Sampling

$$P(\theta \mid D) = \frac{\boxed{P(a|s,\theta)} \cdot P(\theta)}{P(a \mid s)}$$

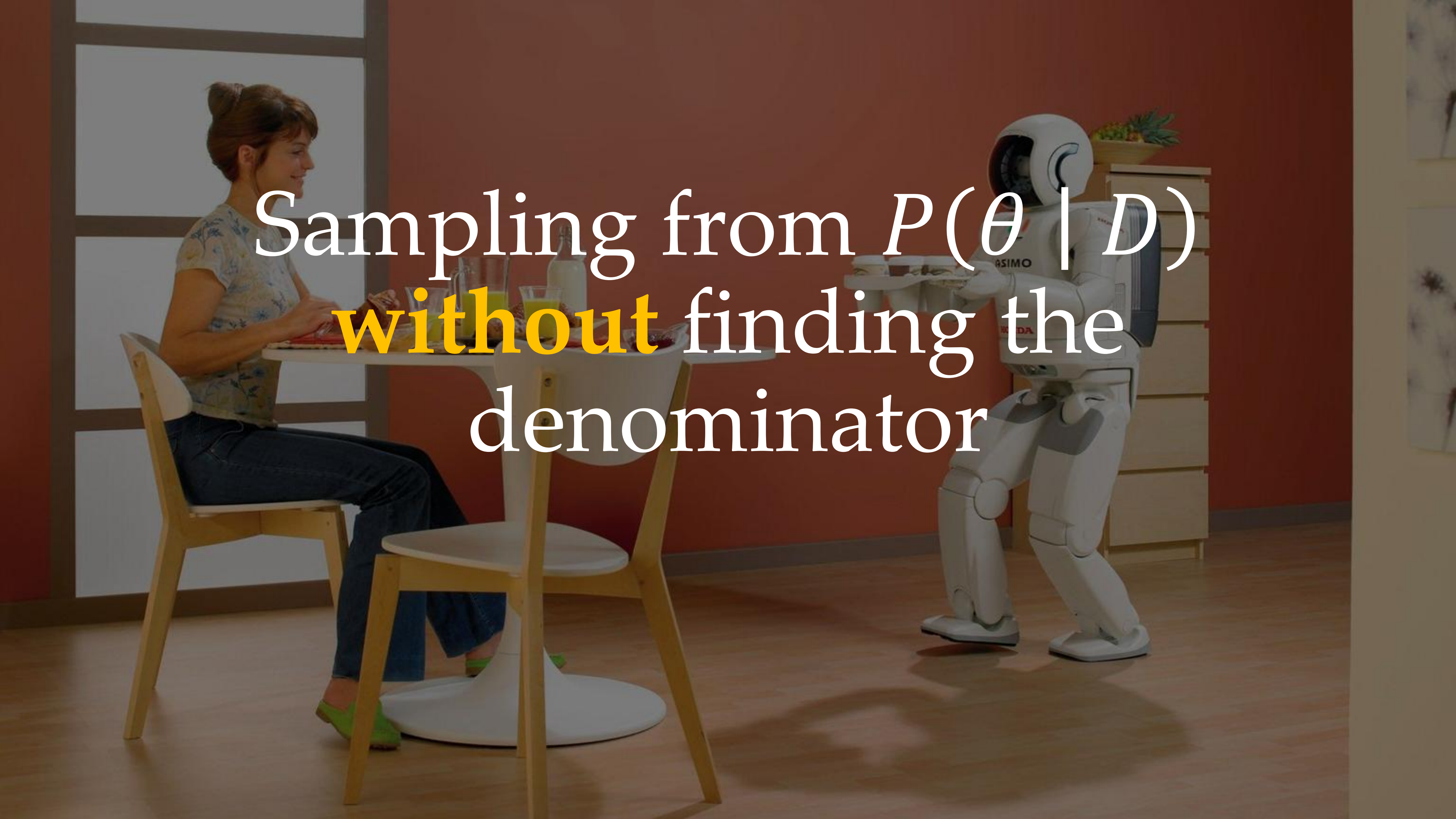
$$P(a \mid s, \theta) = \frac{e^{\beta \cdot Q_\theta(s,a)}}{\underline{\int e^{\beta \cdot Q_\theta(s,a')}\, da'}} = \frac{e^{\beta \cdot Q_\theta(s,a)}}{Z(\theta)}$$

Normalize across continuous set of actions

Sampling

$$P(\theta \mid D) = \frac{e^{\beta \cdot Q_\theta(s,a)} \cdot P(\theta)}{Z(\theta) \cdot P(a \mid s)}$$

- The **numerator** is easy to compute
- The **denominator** is hard; normalize across continuous a and θ spaces



Sampling from $P(\theta \mid D)$ **without** finding the denominator

Metropolis–Hastings

Say you have two parameter choices in mind: θ_1 and θ_2

Given the demonstration, which θ is more likely?

$$\frac{P(\theta_1 \mid D)}{P(\theta_2 \mid D)}$$

If this ratio is smaller than 1, then θ_2 is more likely.
If this ratio is bigger than 1, then θ_1 is more likely.

Metropolis–Hastings

Say you have two parameter choices in mind: θ_1 and θ_2
Given the demonstration, which θ is more likely?

$$\frac{e^{\beta \cdot Q_{\theta_1}(s,a)} \cdot P(\theta_1)}{Z(\theta_1) \cdot P(a \mid s)} \cdot \frac{Z(\theta_2) \cdot P(a \mid s)}{e^{\beta \cdot Q_{\theta_2}(s,a)} \cdot P(\theta_2)}$$

Plugging in our equation for $P(\theta \mid D)$

Metropolis–Hastings

Say you have two parameter choices in mind: θ_1 and θ_2
Given the demonstration, which θ is more likely?

$$\frac{e^{\beta \cdot Q_{\theta_1}(s,a)} \cdot P(\theta_1)}{\cancel{Z(\theta_1)} \cdot \cancel{P(a \mid s)}} \cdot \frac{\cancel{Z(\theta_2)} \cdot \cancel{P(a \mid s)}}{e^{\beta \cdot Q_{\theta_2}(s,a)} \cdot P(\theta_2)}$$

These two terms are equal and cancel.

These two terms are not necessarily equal,
but a common assumption is that we can cancel.

Metropolis–Hastings

Say you have two parameter choices in mind: θ_1 and θ_2
Given the demonstration, which θ is more likely?

$$\frac{P(\theta_1 \mid D)}{P(\theta_2 \mid D)} \approx \frac{e^{\beta \cdot Q_{\theta_1}(s,a)} \cdot P(\theta_1)}{e^{\beta \cdot Q_{\theta_2}(s,a)} \cdot P(\theta_2)}$$

By dividing these terms, we **got rid of the normalizers**!

Metropolis–Hastings

Input: Demonstrations D
Output: N samples of θ from $P(\theta \mid D)$

$$\theta_{samples} = \{\theta^1, \theta^2, \ldots \theta^N\}$$

Metropolis–Hastings

Input: Demonstrations D
Output: N samples of θ from $P(\theta \mid D)$

$\theta \leftarrow$ random choice
$\theta_{samples} \leftarrow \{ \}$
For N iterations:
  append θ to $\theta_{samples}$
  $\theta' \leftarrow$ random choice close to θ
  $\rho \leftarrow$ random number $[0, 1]$
  If $P(\theta' \mid D)/P(\theta \mid D) > \rho$:
    $\theta \leftarrow \theta'$
Return $\theta_{samples}$

$$\frac{P(\theta' \mid D)}{P(\theta \mid D)} \approx \frac{e^{\beta \cdot Q_{\theta'}(s,a)} \cdot P(\theta')}{e^{\beta \cdot Q_{\theta}(s,a)} \cdot P(\theta)}$$

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1. How do we infer a human's reward from demonstrations?
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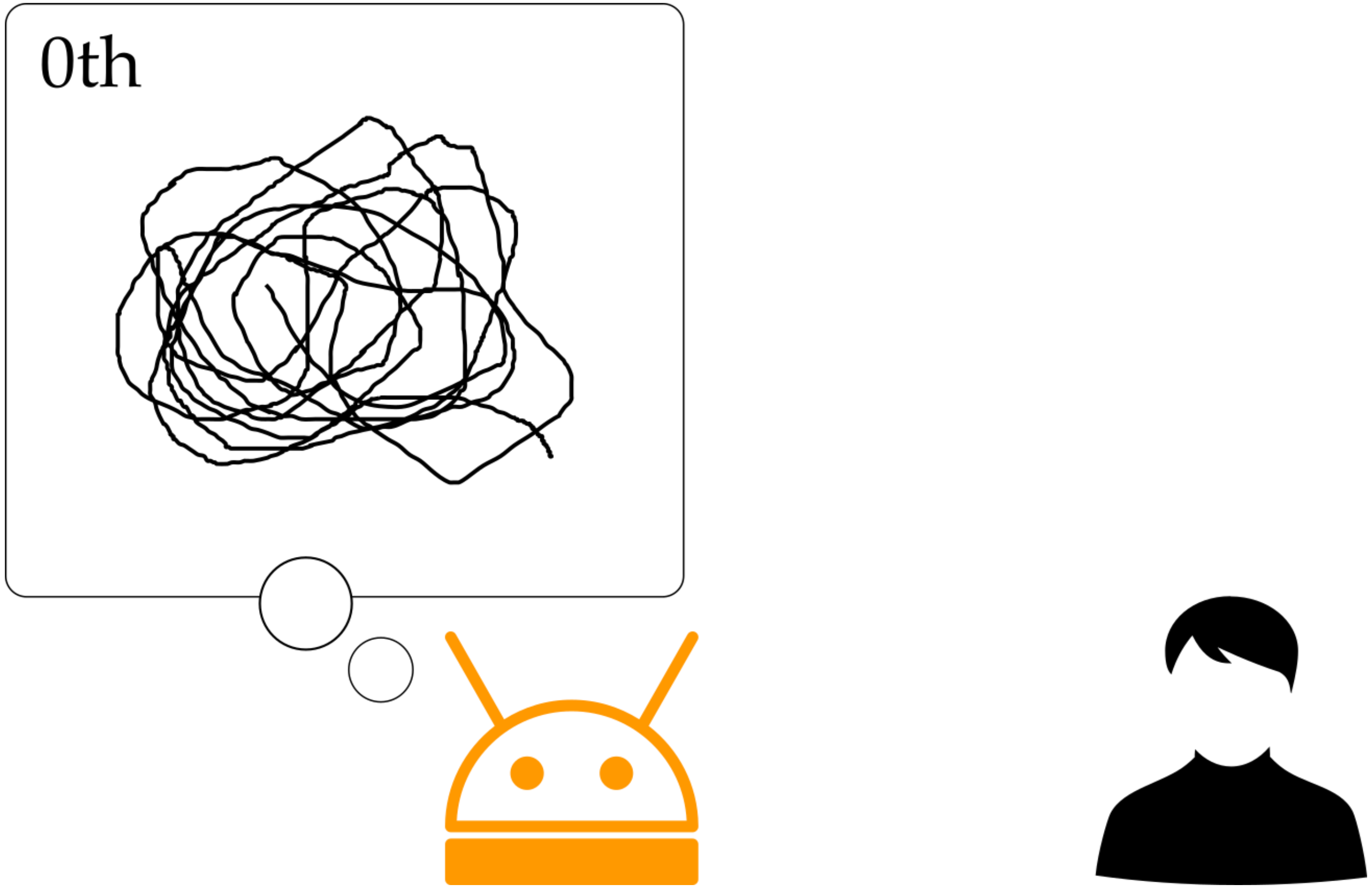
Theory of Mind

Understand other agents by ascribing mental states to them (guessing what is happening in their mind).

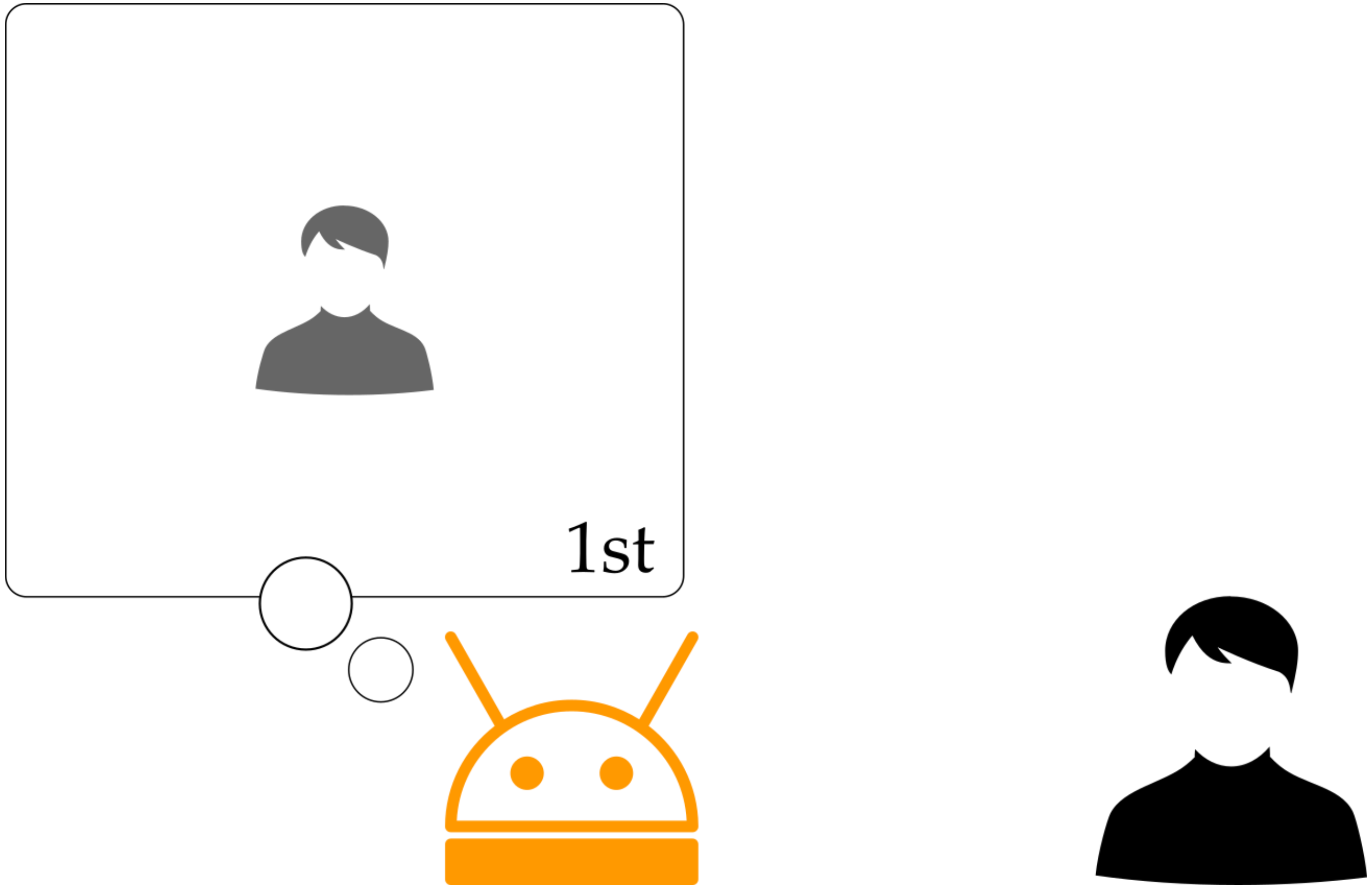
Theory of Mind

What do I think?



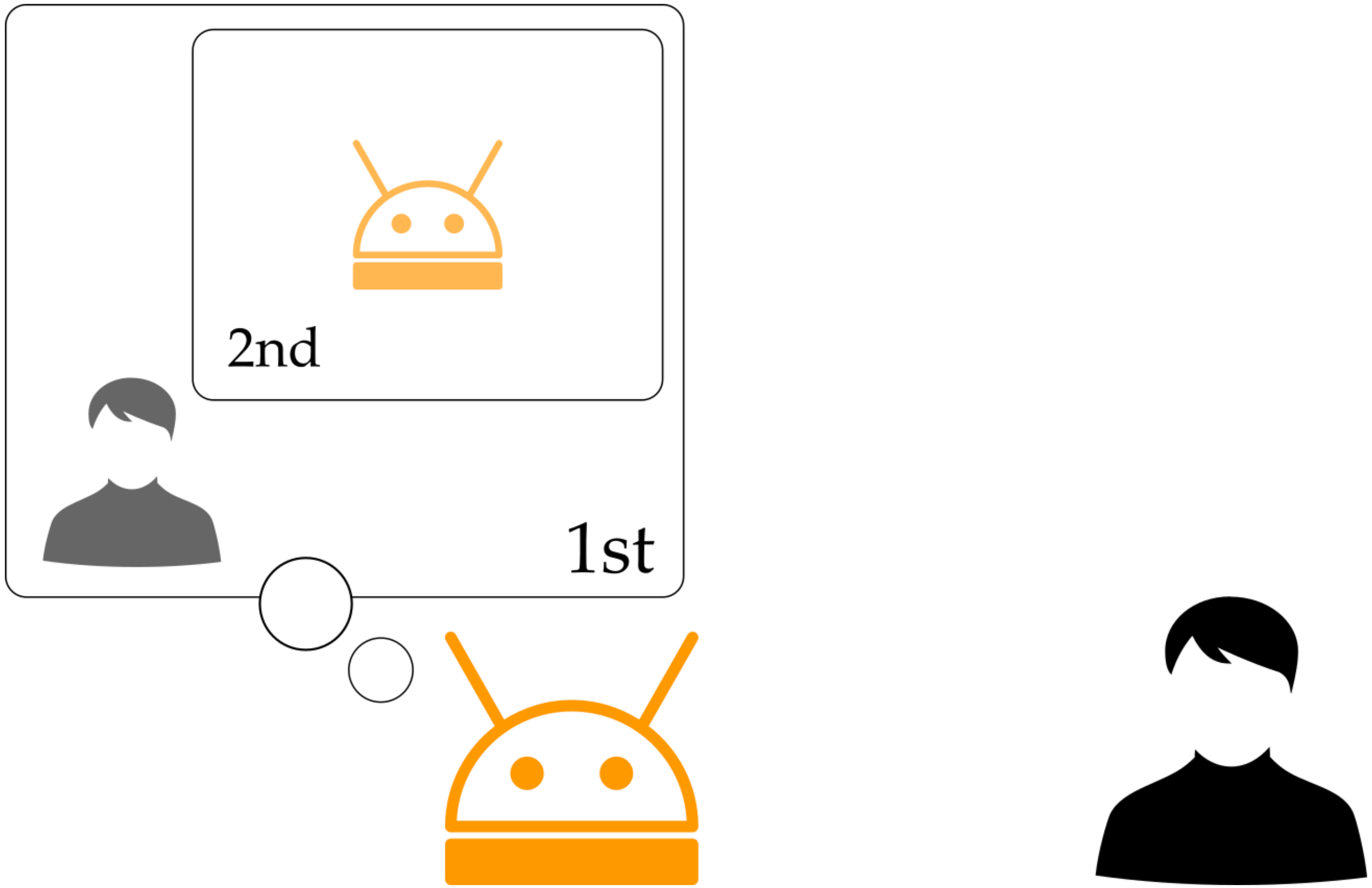
Theory of Mind

What do I think you think?

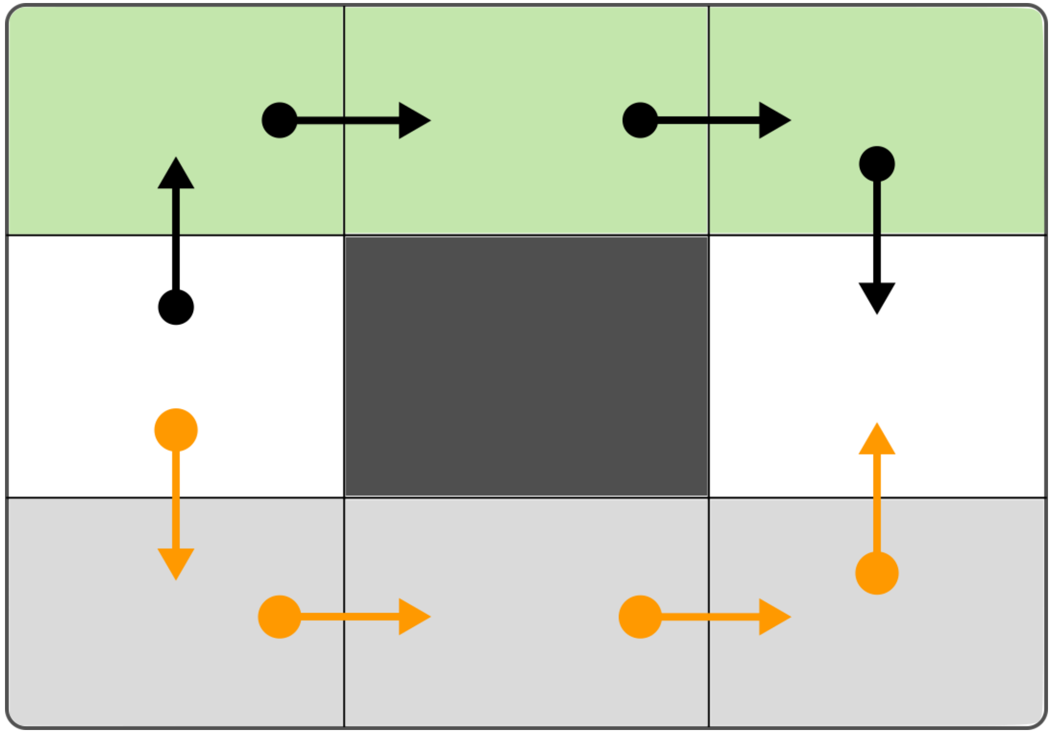


Theory of Mind

What do I think you think I think?

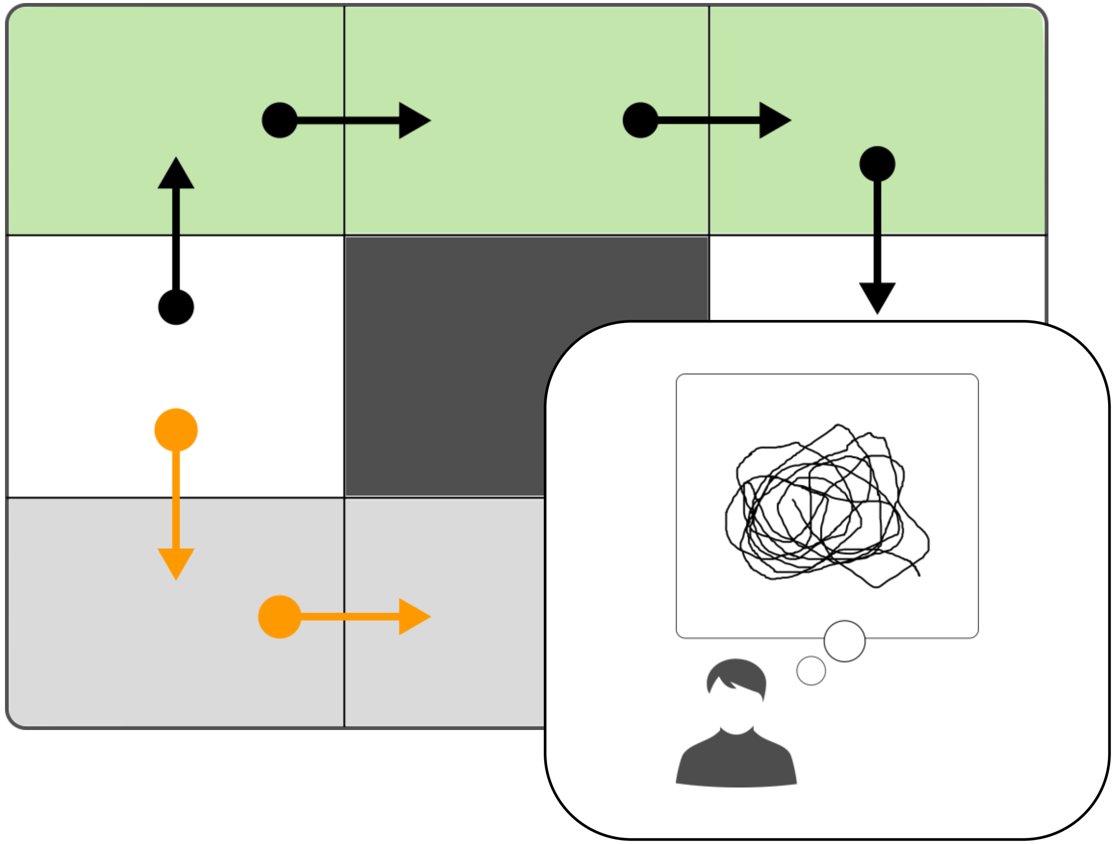


Motivating Problem



- We are trying to walk around an obstacle (middle)
- We do not know if its fine to go over the grass, or if we need to stay on the sidewalk

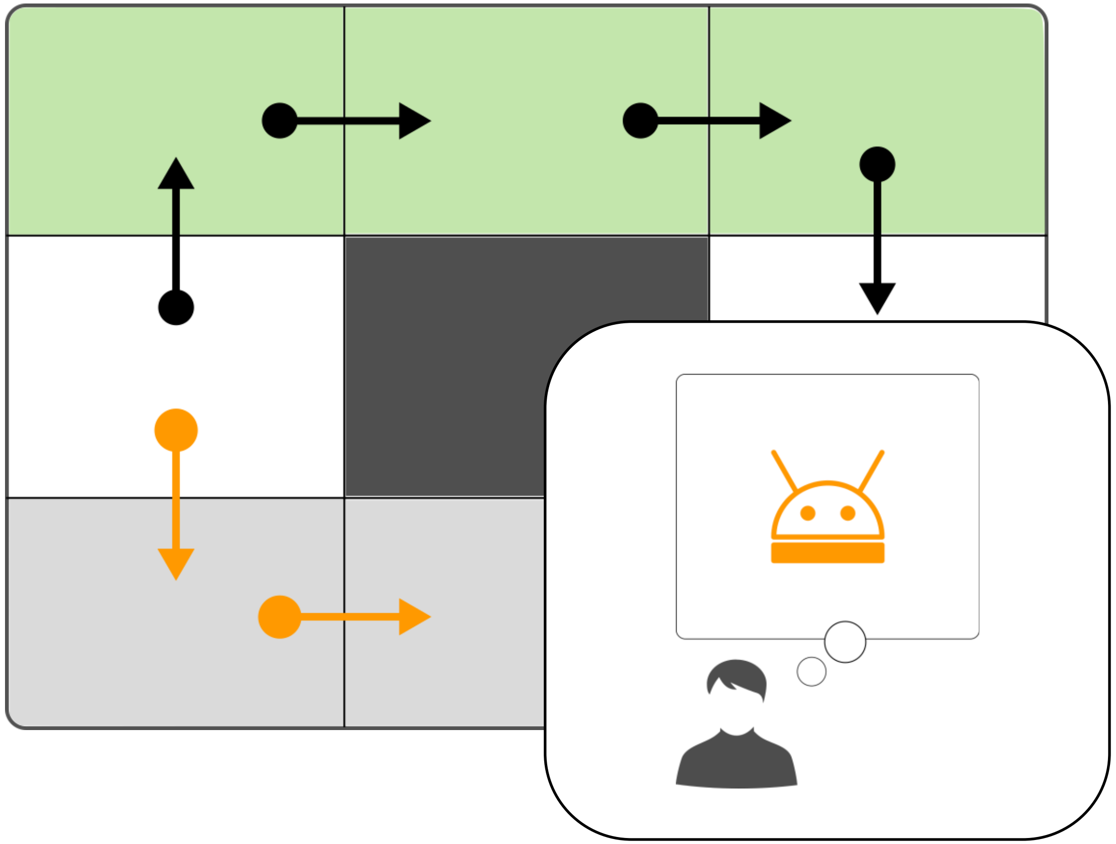
Motivating Problem



- We are trying to walk around an obstacle (middle)
- We do not know if its fine to go over the grass, or if we need to stay on the sidewalk

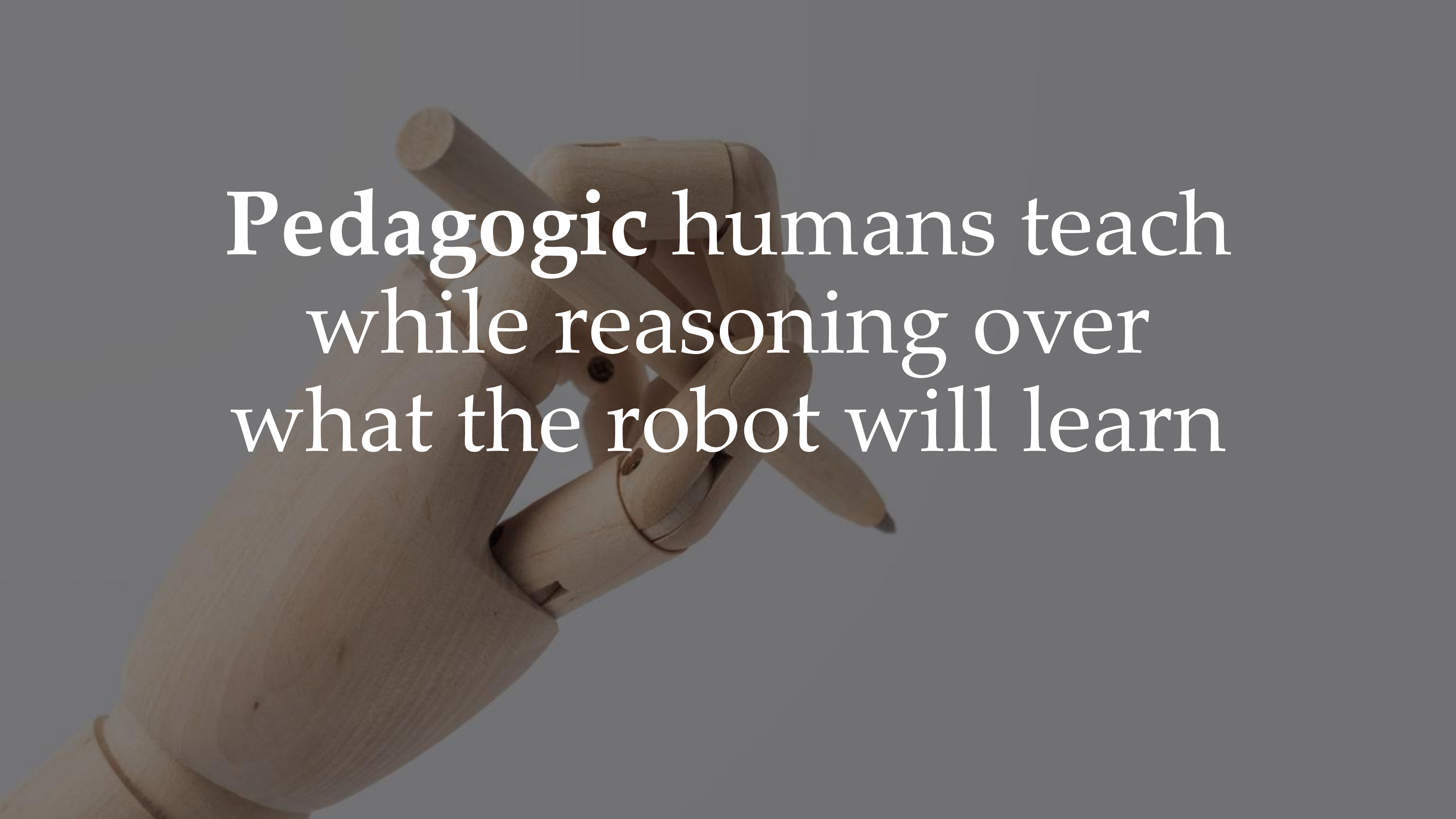
We learn by observing the human. What action would you expect a 0th order human to take?

Motivating Problem



- We are trying to walk around an obstacle (middle)
- We do not know if its fine to go over the grass, or if we need to stay on the sidewalk

We learn by observing the human. What action would you expect a 1st order human to take?



Pedagogic humans teach while reasoning over what the robot will learn

Pedagogic Human

Human with 1st order theory of mind (thinks about robot).

The human knows the task with parameters θ.

What trajectory ξ should they show to best reveal θ to the robot learner?

Pedagogic Human

Human with 1st order theory of mind (thinks about robot).

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What trajectory ξ should they show to best reveal θ to the robot learner?

$$\xi^* = \underset{\xi \in \Xi}{\operatorname{argmax}} \, P(\, \theta \mid \xi \,)$$

Human's model of how the robot learns

This Lecture

- Learning and optimizing rewards in robotics
- What is inverse reinforcement learning?
- Theory of mind

Next Lecture

- Finding efficient state representations.